

Topicalization Defined by Syntax

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28.1 The Syntax and Semantics of Topicalization

Topicalization is a formal mechanism to mark a special kind of subject-predicate relationship, namely, the predicative relation between a topic and its comment along the line of William (1980), as illustrated below:

- (1) $\text{Topic}_k, [\text{Comment} \dots [e]_k \dots]$

The comment part is typically an open clausal predicate, that is, a clause with a gap $[e]$ whose identity can be traced back to the topic (as indicated by the co-indexing $\langle k \rangle$; cf. Shi 2000). The dependency between the topic and the gap is nonargumental in nature (i.e., an A'-dependency), which can be created either by movement, as in (2a), or by pronominal binding, as in (2b) (*pro*: empty pronoun):

- (2) a. $\text{Topic}_k, [\text{Comment} \dots t_k \dots]$
 b. $\text{Topic}_k, [\text{Comment} \dots \text{pro}_k \dots]$

On the other hand, the topic part can be regarded as the gateway from syntax to discourse, where information structure comes into play in sentence formation. Based on a variety of typological considerations, Huang (1984) proposes that even a topic can be empty in Chinese, which serves as a formal medium to access discourse information (e.g., a salient antecedent in the discourse for the empty category $[e]$). As a result, a clear distinction can be drawn between Chinese-type and Romance-type *pro*-drop: empty pronouns are identified by morphological agreements on Infl in Romance languages (cf. Rizzi 1990), whereas they are licensed by a base-generated (null) topic in Chinese languages (cf. Huang 1989).

In fact, even in a non-*pro*-drop language like English, there are two types of topic construals (cf. Cinque 1990; Haegeman 2004, 2012; among many others): one involves movement, as in (3a), and the other not, as is the case of pronominal resumption in (3b):

- (3) a. John_k, I like t_k very much.
 b. John_k, I like him_k very much.

In Chinese, we can easily find a parallel to the English pair (cf. Huang 1982; Xu and Langendoen 1985; Shyu 1995, 2014; Li 2000; Huang et al. 2008; Tsai 2008, among others), as exemplified below:

- (4) a. 阿 Q，我很討厭。
 Akiu_k, __wo__hen__taoyan__t_k.
 Akiu __I__very__hate
 Akiu, I hate very much.
 b. 阿 Q，我很討厭他。
 Akiu_k, __wo__hen__taoyan__ta_k.
 Akiu __I__very__hate__him
 Akiu, I hate him very much.

The difference is that Chinese also allows pro-drop, where an empty pronoun can be employed even when a topic is not visible within the sentence domain, as illustrated below:

- (5) Context: The speaker and the addressee are waiting for someone whose reference is salient in the relevant context/discourse.

Pro-drop: 來了！看到了！

pro lai__le! __*pro*__kan-dao-le__*pro*!¹
 come __Inc__see-reach-Prf
 He is coming! I have seen him!

Furthermore, the base-generation (or external merger in the Minimalist terms) of a topic opens the avenue for a variety of topic-prominent constructions, including the double subject construction in (6a) (cf. Chao 1968), as well as the topic chain construction in (6b) (cf. Tsao 1977):

- (6) a. 阿Q眼睛大。
 Akiu,__yanjing__da.
 Akiu__eye__big
 Akiu has big eyes.
 b. 趙媽，眼睛大，鼻子挺，嘴巴小。
 Zhaoma,__[yanjing__da], __[bizi__ting], __[zuiba__xiao].
 Zhaoma__eye__big__nose__upright__mouth__small
 Akiu has big eyes.

In addition, Shi (2000) carries the investigation further into the peculiar phenomena known as dangling topics, with a view to illustrating the defining characteristics of Chinese-type topic constructions: The first case

¹ This example is taken from the follow two webpages: <http://ccmcc.nidbox.com/diary/read/8055672>; http://m.guanxin.mobi/m_topic_web_212163_1_30.htm.

is indeed a famous one, as originally discussed in Chao (1968) and Li and Thompson (1976, 1981):

- (7) 那場火，幸虧消防隊來得快。
 Na-chang_huo, _xingkui_xiaofang-dui_lai-de-kuai.
 that-CL_fire, _fortunately_fire-brigade_come-DE-fast
As for that fire, fortunately the fire brigade came quickly, (otherwise it) ...

The second case has to do with a reciprocal-like construal associated with a based-generated topic, as illustrated by the following example taken from Huang (1982):

- (8) 他們，我看你，你看我。
 tamen, _wo_kan_ni, _ni_kan_wo.
 they_I_look_you, _you_look_me
They look at each other.

Thirdly, a double subject sentence like (9a) may undergo a reversal construal such as (9b) (taken from Chen 1996). Take a typical possession structure (9c) as a base for comparison: Here it is the apparent possessee *wujia* ‘stuff.price’ rather than the possessor *Niuyue* ‘New York’ that is topicalized:

- (9) a. 紐約，物價最貴。
 Niuyue, _wujia_zui_gui.
 New.York, _living.cost_most_expensive
New York, (its) living cost is most expensive.
 b. 物價，紐約最貴。
 wujia, _Niuyue_zui_gui.
 living.cost, _New.York_most_expensive
As for living cost, New York is most expensive.
 c. [紐約的物價]最貴。
 [Niuyue_de_wujia]_zui_gui.
 New.York_DE_living.cost_most_expensive
The living cost of New York is most expensive.

The remaining two types involve either measuring by predicate nominals as in (10a), or an affective construal such as (10b) (cf. Li and Thompson 1976).

- (10) a. 那種豆子，一斤二十六塊錢。
 na-zhong_douzi, _yijin_sanshi-kuai_qian.
 that-CL_beans_one-CL thirty-CL_money
That kind of beans [topic], one catty is thirty dollars.
 b. 這件事，你不能光麻煩一個人。
 zhe-jian_shiqing, _ni_bu_neng_guang_mafan_yi-ge_ren.
 this-CL_matter_you_not_can_only_bother_one-CL_person
This matter (topic), you can't just bother one person.

Shi points out that the traditional notion of ‘aboutness’ is too vague to capture the essence of the above construals, and that all the Chinese-type topics mentioned above arguably involve a (subject) position inside the comment clause, which may well be subject to syntactic construals such as control or movement à la Huang (1984, 1989).

There are also some objections raised by Pan and Hu (2002) and Xu and Liu 徐烈炯, 劉丹青 (2007) as to this line of thinking. They argued that this particular treatment of dangling topics fail to capture certain generalizations about the argument structure and the predication relationship involved in such construals.

28.2 What Is Topic-Prominent? A Syntactic View

As for the syntax of topic from a formalist view, it is important to note Chomsky’s (1977) seminal proposal that topicalization should be treated on a par with question formation in that both operations involve *wh*-movement. More specifically, what is moved in topicalization is not a *wh*-phrase but a null operator, as represented by *Op* in the following diagram:

- (11) [Discourse ... Topic, [Comment Op_k [... e_k ...]] ...]

Along this line, a topic sentence such as (12a) can be analyzed as (12b), where the null operator has no reference, and needs to be identified (cf. Chomsky 1986):

- (12) a. That book, I like very much.
b. [that book, [Op_k [I like e_k very much]]]

It is worth mentioning that the null operator movement creates an A’-dependency (nonargument dependency) very similar to the case with relativization where there is no *wh*-phrase involved, as (13a) can be analyzed as (13b):

- (13) a. the book (that) I like very much
b. [the book [Op_k (that) [I like e_k very much]]]

The difference is that topicalization triggers syntactic predication (hence the topic-comment relationship, cf. Williams 1980), whereas relativization triggers modification (hence the modifier–head relationship). In terms of semantics, the null operator can translate into a λ -operator that predicates the open comment clause (i.e., a clausal predicate) upon the topic in question, as illustrated below:

- (14) λ_x (... x ...) (Topic)

In fact, Chomsky’s treatment of topicalization also creates the need for an independent topic position in the upper layer of sentence structure,

may well be seen as bound by a definiteness operator expressing topicality (a D-operator in Tsai’s terms), as schematized in (18b):

- (18) a. 剛才人來了。
gangcai__ren__ lai__le. [bare nominal]
just__person__come__Inc
The person has just come.
b. D-operator binding for Chinese bare nominals:
D_x [... N(x) ...]

This way a coherent picture emerges as to how A’-dependencies can be built without movement in licensing topicalization, pro-drop, *wh*-in-situ, and bare nominals. Along this line, the dangling topic construals discussed by Shi also receive a natural account based on the formal mechanism presented above.

Paul and Whitman (2017), on the other hand, point out some defining properties of topic prominence in the literature, as listed below:

- I Obligatoriness of topics
- II Null topics
- III Overt spell-out of the head of TopicP
- IV *Wh* in situ
- V Obligatory topicalization of generic DPs
- VI Multiple topics
- VII Free ordering of argument and adjunct topics

They argue that the traditional views based on the notion of ‘whole language typology’ have certain shortcomings in both descriptive and theoretical terms (cf. Li and Thompson 1976; Hale et al. 1977), and that a proper assortment of the above features might be the right way to go. Interestingly enough, they add different types of language for comparison as to how they fare against the above syntactic criteria in root clauses. The results are illustrated in the following table (cf. Paul and Whitman 2017):

		Chinese	Japanese	German	English
I	Obligatory topics	no	yes	yes	no
II	Null topic	yes	yes	yes	no
III	Overt spell-out of Top°	optional	yes	yes	no
IV	<i>Wh</i> in situ	yes	yes	no	no
V	Generics obligatorily topicalized	no	yes	no	no
VI	Multiple topics	yes	yes	yes(?)	yes(?)
VII	Adjunct/argument topic order free	yes	yes	no?	no?
VIII	Top° filled by internal merge	no	no	yes	yes

The last criterion is added to cut across the group of languages: Chinese and Japanese have Top° filled by an overt topic particle, whereas German and English have a tensed verb raised to Top° as in (residual) V2 phenomena. For instance, the following Japanese topic sentence requires an overt topic marker *wa* (Tateishi 2006: 65):

- (19) Hannin *wa*/**ga* Saburoo *ga* *ayasii*.
 Perpetrator__Top/Nom__Saburoo Nom__suspicious
As for the perpetrator, Saburoo is suspicious.

Its Chinese counterpart, on the other hand, allows the topic marker *a* to appear optionally:

- (20) (zhiyu) fanren (a) Akiu zui *keyi*.
 as.for_perpetrator_Top_Akiu_most_suspicious
As for the perpetrator, Akiu is most suspicious.

By contrast, the topic head is arguably filled with (tensed) verb raising in German (hence typical V2 phenomena in main clauses, cf. den Besten 1983; Grewendorf 2012, among others). This can be easily seen with the positioning of *habe* ‘have’ behind the topic *Dieses Gerücht* ‘this rumor’ in (21):

- (21) [*Dieses Gerücht* [**habe**] [*ich von meiner Sekretärin gehört*]].
 this__Acc__rumor__have__1SG__from__my__secretary__heard
I heard this rumor from my secretary.

Although English does not display full-fledged V2 properties, it still has some residual construals to show for, which are often collectively known as ‘inversion’ phenomena:

- (22) a. [*Seldom* [**have**] [*I met a politician like him*]].
 b. [*Nowhere* [**have**] [*I had such bad luck*]].
 c. [*Where* [**have**] [*you been*]]?

Other test results show similarities shared by some of the languages, but not the others. For example, a generic subject must be topicalized in Japanese, as in (23a), while a definite NP can be associated with either the topic marker *wa* or the subject marker *ga*, as in (23b).

- (23) a. Inu *wa* [*honyuu doobutu da*]. (categorical judgment)
 dog__Top__mammal__is
The dog is a mammal.
 b. Poti *wa/ga* [*hone o kazitte iru*]. (thetic judgment)
 Pochi__Top/Nom__bone ACCchewing__is
Pochi is chewing a bone.

This syntactic contrast of categorical vs. thematic judgments in Kuroda’s (1988) sense is not attested in Chinese, German, and English. Furthermore, although the four languages allow multiple topics in the

sentence-initial position to various degrees,² only Chinese (and Japanese) allow a free order change of adjunct and argument topics, as exemplified by the following minimal pair:

- (24) 阿 Q, 昨天, 我給了他一本書。
 Akiu, __zuotian, __wo __gei-le __ta __yi-ben __shu.
 Akiu_yesterday __I __give-Prf __himone-Cl __book
 Akiu, yesterday, I gave him a book.
- (25) 昨天, 阿 Q, 我給了他一本書。
 zuotian, __Akiu, __wo __gei-le __ta __yi-ben __shu.
 yesterday __Akiu, __I __give-Prf __him __one-Cl __book
 Akiu, yesterday, I gave him a book.

As a matter of fact, the (partial) correlations of these typological features fall right under the recent debates on whether we should embrace the notion of macro parameters or that of micro parameters when generalizations are to be extracted from systematic variations among languages (cf. Borer 1984; Newmeyer 2005; Baker 2008; Biberauer 2008; Haspelmath 2008; Huang and Roberts 2017, among others).

28.3 Information Structure and Interface Economy

In the literature, there are often concerns about whether we should incorporate various semantic/pragmatic considerations into the computation of core syntax, and how the notion of information structure should be encoded in morphosyntactic terms (see, for example, Shyu 2016). Rizzi (1997) and Cinque (1999) provide an explicit and positive solution by ‘syntacticizing’ those considerations in terms of functional features, which in turn form the backbone of the left periphery (i.e., the CP domain). Rizzi, in particular, proposes to distinguish two major systems of the complementizer layer:

- I The Force-Finiteness System, where the Force head encodes speech acts, and illocutionary force, where the Fin(iteness) head encodes the finite/nonfinite distinction, as well as the realis/irrealis mood.

² Typical examples of multiple topics can be found in both Chinese and Japanese, as shown below respectively (cf. Paul and Whitman 2017). Similar construals are much more restricted in German and English.

- (i) 中國, 大城市, 上海, 交通比較亂。
 Zhongguo, __da __chengshi, __Shanghai, __jiaotong __bijiao __luan.
 China, __big __city, __Shanghai __traffic __relatively __chaotic
 As for China, as for big towns, Shanghai, the traffic is rather chaotic.
- (ii) Tyuugoku wa daitokai wa Syanhai wa koutuu ga midarete iru.
 China __Top __big __city __Top __Shanghai __Top __traffic __Nom __disordered __is
 As for China, as for big towns, Shanghai, the traffic is chaotic.

- II The Topic-Focus System, where he distinguishes two types of topic constructions: one involves clitic left dislocation (CLLD) construals (cf. Section 28.4), whereas the other involves null operator construals.

Furthermore, it is instructive to note the following asymmetries between a topic and a focus (cf. Rizzi 1997: 286–291):

- III Topics are identified through either cliticization or null operator binding. Topicalization displays no weak crossover effects, as in (26), and is compatible with a fronted *wh* operator (as well as other topics), as in (27):

- (26) 阿 Q_k (啊), [他_k的爺爺]很寵他_k。
 Akiu_k (a), [ta_k de yeye] hen chong ta.
 Akiu_{Top}, he DE grandfather very pamper him
 Akiu_k, his_k grandfather pampered him.
- (27) 阿 Q_k (啊), 什麼_j, 他_k最愛吃 t_j?
 Akiu_k (a), shenme_j, ta_k zui ai-chi t_j?
 Akiu_{Top}, what, he most love-eat
 Akiu_k, what does he_k love to eat?

The relevant syntactic configuration is sketched as follows:

- (28)
-
- ```

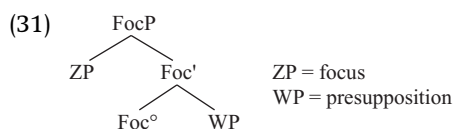
graph TD
 TopP --> XP
 TopP --> TopPrime[Top']
 TopPrime --> TopDegree[Top°]
 TopPrime --> YP

```
- XP = topic  
YP = comment

- IV Foci, on the other hand, involve quantification through operator-variable binding. Focalization typically displays weak crossover effects, as in (29), while utilizing a unique syntactic position. Moreover, a focus is not compatible with a fronted *wh* operator, as in (30):

- (29) \* 阿 Q<sub>k</sub>, [他<sub>k</sub>的爺爺]很寵 t<sub>k</sub>。小 D<sub>j</sub>, [他<sub>j</sub>的爺爺]也很寵 t<sub>j</sub>。  
 \* Akiu<sub>k</sub>, [ta<sub>k</sub> de yeye] hen chong t<sub>k</sub>. Xiaodi<sub>j</sub>, [ta<sub>j</sub> de yeye] ye hen chong t<sub>j</sub>.  
 Akiu, he DE grandfather very pamper t<sub>k</sub>. Xiaodi, he DE grandfather also very pamper.  
 \* Akiu<sub>k</sub>, his<sub>k</sub> grandfather pampered t<sub>k</sub>. Xiaodi<sub>j</sub>, his<sub>j</sub> grandfather also pampered t<sub>j</sub>.
- (30) \* 阿 Q<sub>k</sub>, 什麼<sub>j</sub>, t<sub>k</sub>最愛吃 t<sub>j</sub>? 小 D<sub>j</sub>, 什麼<sub>h</sub>, t<sub>j</sub>又最愛吃 t<sub>h</sub>?  
 \* Akiu<sub>k</sub>, shenme<sub>j</sub>, t<sub>k</sub> zui ai-chi t<sub>j</sub>? Xiaodi<sub>j</sub>, shenme<sub>h</sub> t<sub>j</sub> you zui ai-chi t<sub>h</sub>?  
 Akiu, what most love-eat? Xiaodi<sub>j</sub>, what on.the.other hand most love-eat?  
 \* Akiu<sub>k</sub>, what<sub>j</sub> does he<sub>k</sub> love to eat t<sub>j</sub>? Xiaodi<sub>j</sub>, what<sub>h</sub> does he<sub>j</sub> love to eat t<sub>h</sub> on the other hand?

The relevant syntactic configuration is sketched as follows:



With this development of new technology, one may wonder if there is a principle regulating the rich inventory of ‘syntacticized’ concepts, and how this approach may fare with the core values of the Minimalist Program as laid out in Chomsky (1995). One way to think of this issue is to build upon Reinhart’s (1996) notion of interface economy: in essence, topicalization and focus fronting apply only to serve the interpretive needs on the interface, be it PF (imposed by the perceptual-articulatory system) or LF (imposed by the information structure/the conceptual-intentional system). This move in turn justifies the introduction of various functional heads (including  $\text{Top}^\circ$  and  $\text{Foc}^\circ$ ) in the top layer of a sentence, which would otherwise be unavailable under the traditional view of the autonomy of syntax. We will explore this line of thinking further with more concrete data in the next section.

## 28.4 The Topography of Chinese Topics

Since the seminal work of Rizzi (1997, 2004), the functional structure of CP has received much fine-grained and articulate analysis, and the basic ‘topography’ of the topic field has been laid out as follows (Force: illocutionary force head; Int: interrogative head; Foc: focus head; Fin: finiteness head):

(32) Force ( $\text{Top}^*$ ) Int ( $\text{Top}^*$ ) Foc ( $\text{Top}^*$ ) Fin IP

Nevertheless, the distribution of topics depicted in (26) is both too simple and too complicated, partly because it says nothing about the hierarchical arrangement of different types of topics across languages, partly because there seem to be some linguistic variations which call for further adjustment of the above picture. This consideration leads us to the work of Badan and Del Gobbo (2011), where it has been proposed that distinct types of Chinese topics occupy distinct types of syntactic positions. Their relative word order can be spelled out in the following cartographic terms:<sup>3</sup>

(33) Aboutness Topic > Hanging Topic (HT) > Left Dislocation (LD)<sup>4</sup> > *lian*-Focus > IP

<sup>3</sup> Alternatively, one can find a more basic view of the left periphery of Mandarin Chinese in Paul (2005):

(i) CP(force) > TopicP > *even* Focus > IP > ...

<sup>4</sup> The term ‘left dislocation’ is traditionally used to refer to nonmovement topicalization (cf. Cinque 1990). There is no such connotation in Badan and Del Gobbo’s (2011, 2015) terminology.

Naturally, it is important to carry out more tests to substantiate this proposed topography of Chinese topics. First of all, the placement of an aboutness topic is relatively uncontroversial, as it is based-generated high above in the left periphery, typically expressing relations such as ‘whole–part’ and ‘set–member’. As illustrated below, the aboutness topic *jiaren-zhong* ‘family-in’ stands higher than the HT *xiaodi* ‘little brother’ but not the other way around, as evidenced by the contrast between (34a,b)

- (34) a. 家人中，小弟，我對他最好。  
           *jiaren-zhong, xiao.di, wo dui ta zui hao.*  
           family-in little.brother, I to him most good  
           *In the family, little brother, I am best to him.*  
           (Aboutness Topic > HT)
- b. \* 小弟，家人中，我對他最好。  
       \* *xiao.di, jiaren-zhong, wo dui ta zui hao.*  
       little.brother, family-in, I to him most good  
       (\*HT > Aboutness Topic)

The defining properties of HT are given in (35a–e), which present a clear contrast to those of LD, as in (36a–e) (see also Kuong 2006):

- (35) a. HT can only be a Bare DP (it cannot be a PP).  
       b. It does not need to agree in Case with the coreferent resumptive element inside the predicate (IP).  
       c. HTs always require a resumptive pronoun expressing the type of argument: it only agrees with the HT in number and gender, not in Case.  
       d. It can be resumed by an epithet or a tonic pronoun.  
       e. Multiple HTs are not possible.
- (36) a. LD can be a PP.  
       b. It agrees in Case with the coreferent resumptive pronoun inside the predicate (IP).  
       c. LD elements require a resumptive pronoun (clitic, in Italian) only when they correspond to direct or partitive objects.  
       d. It cannot be resumed by an epithet or a tonic pronoun.  
       e. Multiple LDs are possible.

As exemplified below, the aboutness topic again positions itself higher than the HT, which in turn scopes over the LD *cong xuexiao* ‘from school’. This is illustrated by the contrast between (37) on the one hand and (37b–f) on the other:

- (37) a. 家人中，小弟，從學校，我常帶他回家。  
           *jiaren-zhong, xiao.di, cong xuexiao, wo chang dai ta hui jia.*  
           family-in, little.brother, from school I often bring him back home  
           *In the family, little brother, from school, I often brought him back home.*  
           (Aboutness Topic > HT > LD)

- b. \* 小弟，家人中，從學校，我常帶他回家。  
 \* xiao.di, \_\_jiaren-zhong, \_\_cong\_xuexiao, \_\_wo\_chang\_dai\_\_  
 ta\_hui\_jia.  
 little.brother, \_\_family-in, \_\_from\_school\_I\_often\_\_  
 bring\_him\_back\_home  
 (\*HT > Aboutness Topic > LD)
- c. \* 小弟，從學校，家人中，我常帶他回家。  
 \* xiao.di, \_\_cong\_xuexiao, \_\_jiaren-zhong, \_\_wo\_chang\_dai\_\_  
 ta\_hui\_jia.  
 little.brother\_\_from\_school,\_\_family-in,\_\_I\_often\_bring\_\_  
 him\_back\_home  
 (\*HT > LD > Aboutness Topic)
- d. \* 從學校，小弟，家人中，我常帶他回家。  
 \* cong\_xuexiao, \_\_xiao.di, \_\_jiaren-zhong, \_\_wo\_chang\_\_  
 dai\_ta\_hui\_jia.  
 from school little.brother family-in I often bring him  
 back home  
 (\*LD > HT > Aboutness Topic)
- e. \* 從學校，家人中，小弟，我常帶他回家。  
 \* cong\_xuexiao, \_\_jiaren-zhong, \_\_xiao.di, \_\_wo\_chang\_\_  
 dai\_ta\_hui\_jia.  
 From\_school\_family-in,\_\_little.brother\_I\_often\_\_  
 bring\_him\_back\_home  
 (\*LD > Aboutness Topic > HT)
- f. \* 家人中，從學校，小弟，我常帶他回家。  
 \* jiaren-zhong, \_\_cong\_xuexiao, \_\_xiao.di, \_\_wo\_chang\_\_  
 dai\_ta\_hui\_jia.  
 family-in\_\_from\_school\_\_little.brother\_I\_often\_\_  
 bring\_him\_back\_home  
 (\*Aboutness Topic > LD > HT)

The lower part of the topic field is occupied by the *lian*-phrase, whose theoretical status was under much debate in the literature (cf. Shyu 1995, 2014; Paul 2005; Badan and Del Gobbo 2015; Cheung 2015, among others). It is fair to say that it combines both topicality and focus/scalar-sensitive semantics. Syntax-wise, the aboutness topic again stands higher than the *lian*-Focus, as shown below (examples taken from Badan and Del Gobbo 2011):

- (38) a. 花，連玫瑰花，都很便宜。  
 hua, \_\_lian\_meiguihua<sub>k</sub>, \_\_t<sub>k</sub>\_dou\_hen\_pianyi.  
 flowers\_even\_roses\_all\_very\_cheap  
 As for flowers, even roses are cheap.  
 (Aboutness Topic > *lian*-Focus)
- b. \* 連玫瑰花，花，都很便宜。  
 \* lian\_meiguihua<sub>k</sub>, \_\_hua, \_\_t<sub>k</sub>\_dou\_hen\_pianyi.  
 even\_roses\_flowers\_all\_very\_cheap  
 As for flowers, even roses are cheap.  
 (\**lian*-Focus > Aboutness Topic)

Similarly, both the HT and the LD scopes over the *lian*-Focus, as evidenced by the contrast of (39a,b) and that of (40a,b), respectively (examples taken from Badan and Del Gobbo 2011):

- (39) a. 小瑜，連那本很貴的書，李四都給他買了。  
 Xiaoyu<sub>i</sub>, *lian* *na* *ben* *hen* *gui* *de* *shu*<sub>k</sub>, *Lisi* *dou* *gei* *ta*<sub>i</sub> *mai-le* *t*<sub>k</sub>.  
 Xiaoyu, *even* *that-Cl* *very* *expensive* *DE* *book* *Lisi* *all* *to* *her* *buy-Prf*  
*Xiaoyu, even that expensive book, Lisi bought for her.*  
 (HT > *lian*-Focus)
- b. \* 連那本很貴的書，小瑜，李四都給他買了。  
 \* *lian* *na* *ben* *hen* *gui* *de* *shu*<sub>k</sub>, Xiaoyu<sub>i</sub>, *Lisi* *dou* *gei* *ta*<sub>i</sub> *mai-le* *t*<sub>k</sub>.  
*even* *that-Cl* *very* *expensive* *DE* *book* *Xiaoyu*, *Lisi* *all* *to* *her* *buy-Prf*  
*Xiaoyu, even that expensive book, Lisi bought for her.*  
 (\**lian*-Focus > HT)
- (40) a. 給小瑜，連那本很貴的書，李四都買了。  
*gei* *Xiaoyu*<sub>i</sub>, *lian* *na* *ben* *hen* *gui* *de* *shu*<sub>k</sub>, *Lisi* *dou* *mai-le* *t*<sub>k</sub>.  
*to* *Xiaoyu*, *even* *that-Cl* *very* *expensive* *DE* *book*, *Lisi* *all* *buy-Prf*  
*Xiaoyu, even that expensive book, Lisi bought for her.*  
 (LD > *lian*-Focus)
- b. \* 連那本很貴的書，給小瑜，李四都買了。  
 \* *lian* *na* *ben* *hen* *gui* *de* *shu*<sub>k</sub>, *gei* *Xiaoyu*<sub>i</sub>, *Lisi* *dou* *mai-le* *t*<sub>k</sub>.  
*even* *that-Cl* *very* *expensive* *DE* *book*, *to* *Xiaoyu*, *Lisi* *all* *buy-Prf*  
*Xiaoyu, even that expensive book, Lisi bought for her.*  
 (\**lian*-Focus > LD)

The relative order of the aboutness topic, the HT, the LD and the *lian*-Focus is thus fixed in cartographic terms, as correctly predicted by the topic hierarchy (33).

Another case of the manifestation of the topography in question comes from Ai's (2014) treatment of (pseudo-)gapping in Chinese. He presents an alternative to the empty verb analysis of Tang (2001), where IP ellipsis applies at PF only after topicalization and focus movement to the left periphery. Take (41) for example (Ai 2014: 133):

- (41) Q: 那天，在山上，他們都看見了誰？  
*natian*, *zai* *shan-shang*, *tamen* *dou* *kanjian-le* *shei*?  
*that.day* *on* *mountain-above* *they* *all* *see-Prf* *who*  
*(Lit.) That day, on the mountain, they all saw whom?*

- The derivation of the gapping part of the elided answer is illustrated in (42a,b):

- ## 28.5 Noncanonical Variants: Postverbal Topics, Topic Copying, and Reflexive of Nature

(43) a. 自私自利大敗類，滾開吧你！<sup>5</sup>  
 zisizili\_dabailei, \_gun-kai\_ba\_ni! (postverbal topic)  
 selfish\_bastard, \_get-out\_SFP\_you  
*You selfish bastard, get out of here!*

b. 我直接說「你去死吧你！」<sup>6</sup>  
 wo\_zhijie\_shuo “ni\_qu-si\_ba\_ni!” (topic copying)  
 I\_bluntly\_say\_yougo-die\_SFP\_you  
*I said bluntly “you go to hell you!”.*

Even more interestingly, the type of topics can have a double in the sentence-initial position, as evidenced by (43b) (see also Xu and Liu 徐烈炯, 劉丹青 2007, among others). Therefore, we will need to find a fine-grained syntactic analysis which allows large-scale fronting over the topic in question. One way to approach it is to employ the kind of remnant movement proposed by Kayne (1994). As sketched in the following step-by-step derivation of (43a):

- As for the case of topic copying in (43b), the derivation is essentially the same exception that the lower copy of the subject chain ( $ni \dots <ni>$ ) is not deleted at PF, hence the copying effect on the surface, as illustrated below:

<sup>6</sup> From “蘋果日報”, August 20, 2014.

- (45) 
$$\begin{array}{c} [\text{MoodP } \text{ba } [\text{FinP } \text{ni } [\text{vP } <\text{ni}> \text{qu-si}]]] \\ \text{SFP} \quad \text{you} \quad \text{you go-die} \\ (\text{subject raising}) \end{array}$$
- $\Rightarrow$  
$$\begin{array}{c} [\text{FocP } [\text{vP } <\text{ni}> \text{qu-si}]_k \text{ } [\text{MoodP } \text{ba } [\text{FinP } \text{ni } \text{t}_k]]] \\ \uparrow \\ (\text{focus remnant movement}) \end{array}$$

Another intriguing case comes from reflexives of nature (cf. Tsai 2015, 2019). As exemplified by (46a) and (47a), they trigger predication on a cause event in the discourse, thus forming a topic-comment structure on a very large scale. The outer reflexive adverbial *ziji* can then be analyzed as a  $\lambda$ -operator SELF, which binds a Neo-Davidsonian event argument in the tradition of Parsons (1995), as in (46b) and (47b):

- (46) a. 風大了，門自己會開。  
           feng da-le, \_\_men ziji hui kai.  
           wind strong-Inc, \_\_door self will open.  
           *When wind becomes stronger, the door naturally will open.*
- b.  $[\text{TopP } [\text{feng da-le}], \text{SELF}_E\text{-Top}^\circ [\text{men hui kai(E)}]]$   
           wind big-Inc                                   door will open
- c.  $\lambda x \exists e (\text{CAUSE } (x, e) \text{ and } \text{open}(e) \text{ and Theme } (\text{door}, e)) \text{ (stronger wind)}$
- (47) a. 時間到了，花自己會開。  
           Shijian dao-le, \_\_hua ziji hui kai.  
           Time due-Inc, \_\_flower self will blossom.  
           *When the time is due, flowers will blossom by nature.*
- b.  $[\text{TopP } [\text{shijian dao-le}], \text{SELF}_E\text{-Top}^\circ [\text{hua hui kai(E)}]]$   
           time due-Inc                                   flower will blossom
- c.  $\lambda x \exists e (\text{CAUSE } (x, e) \text{ and } \text{blossom}(e) \text{ and Theme } (\text{flower}, e)) \text{ (due time)}$

The topic-comment construal is therefore wired into the causality semantics of an outer reflexive. Along this line, the ‘by nature’ interpretation can be understood as ‘by the nature of a cause–effect relationship’, as represented by the semantic treatments of (46c) and (47c).

## 28.6 Conclusion

In this chapter, we present a general picture of how topicalization and topicality can be defined in morphosyntactic terms. We start with an overview of the syntax and semantics of topics from the vantage point of generative syntax and formal logic. We then show that the notion of topic prominence can be defined by typological correlations through a systematic survey across languages. By applying objective tests to sort out the ‘grey areas’ or the spectrum effects of topicality, this line of research may well contribute to our pursuit of an optimal solution to the mystery of linguistic variations: Both the macro-parameter setting of analyticity and the micro-parameter setting of null topic are to be justified on empirical grounds,

which, if so justified, will provide an explicit account of the inner working of topicalization, as well as its interaction with other types of syntactic mechanisms such as pro-drop and focalization.

Furthermore, the recent advancement of grammatical theories also offers a way to relate the core syntax to language use in the form of syntactic cartography. It becomes clear that the information structure can be encoded by the functional projections in the left periphery, and that various pragmatic/semantic considerations (e.g., aboutness, illocutionary forces, contrastivity, modality, etc.) may enter into the computation of the core syntax by fulfilling the interpretive needs on the interfaces. Through a painstaking investigation of the relative word order of various types of Chinese topics, a general topography has emerged, very much conforming to those observed in Italian and Hungarian. At the end, we test the above approach with a few noncanonical cases such as postverbal topics, topic copying, and reflexives of nature, producing some quite interesting results.

As we enter the age of big data, it is essential to use every tool available in our endeavor to explain the true nature of human language. In this sense, statistics do matter, but they have to be done in a big (and bigger) way. In other words, sometimes we do not see it not because it is rare or does not exist, but because the data simply are not big enough. We need to look at the right place with the right tools, that is, rational reasoning and the precision of the scientific methods such as minimal pairing and experimental studies. It is my belief that the generative power of human language capacity serves to meet the extensional needs through the interfaces; that is, the core syntax provides the backbone of language use, which in turn makes communication possible. As far as I can see, there is no reason why syntactic lab tests cannot be supplemented by naturalistic data and corpus search. After all, open-mindedness and a bit of imagination may go a long way, providing the key to the convergence of different approaches to topicalization and topicality.

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